Advanced IPTV Recording & Management System

Technical Proposal for CMC

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Version 1.8

This proposal supersedes all previous versions of the proposal

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Executive Summary

BesTel is pleased to present this proposal for an Advanced IPTV Recording & Management System, Analysys, designed to capture, store, and manage television content from multiple channels simultaneously. This comprehensive solution will enable BesTel to record up to 32 channels continuously for 30 days, create and catalog video clips, and implement intelligent content detection features. The system is designed with a keen focus on flexibility, scalability, availability and efficiency to suit client's specific requirements.

Analysys System Overview

Our proposed IPTV Recording & Management System is a sophisticated platform that combines hardware infrastructure with advanced software capabilities to deliver a complete solution for television content capture, storage, and management.

Core Capabilities

1. Multi-Channel Recording
2. Long-Term Storage
3. Clip Creation & Management
4. Scheduled Recording
5. Cloud-Based Access (Optional)
6. Comprehensive Search Functionality
7. Analytics Dashboard

Integrated Features of Analysys

1. **Multi-Channel Recording**

   Analysys will simultaneously capture and record content from 32 IPTV channels:

   a. **Continuous 24/7 Recording:** Uninterrupted capture of all broadcast content

   b. **Multiple Format Support:** Compatible with various video codecs and resolutions

   c. **Metadata Capture:** Automatic recording of program information, timestamps, and channel data

   d. **Redundant Recording:** Backup mechanisms to prevent data loss during system failures

   e. **Quality Preservation:** High-fidelity recording maintaining original broadcast quality

   f. **On Demand Recording:** Change recording schedules on the fly

2. **Long-Term Storage**

   Comprehensive storage solution for maintaining a 30-day rolling archive:

   a. **Automated Archive Management:** Intelligent storage system maintaining a 30-day rolling window
   b. **Optimized Compression:** Space-efficient storage without significant quality loss
   c. **Tiered Storage Architecture:** Hot storage for recent content, cold storage for older recordings
   d. **Redundant Array:** RAID configuration to protect against drive failures
   e. **Expandable Design:** Ability to increase storage capacity as needed

3. **Clip Creation & Management**

   Powerful tools for creating, editing, and organizing video segments:

   a. **Intuitive Clip Editor:** User-friendly interface for precise clip creation
   b. **Custom Naming & Tagging:** Flexible organization system for easy retrieval
   c. **Batch Processing:** Create and manage multiple clips simultaneously
   d. **Format Conversion:** Export clips in various formats for different use cases
   e. **Content Library:** Organized repository of all saved clips with preview capabilities

4. **Scheduled Recording**

   Automated recording based on predefined schedules:

   a. **Calendar Interface:** Easy-to-use scheduling tool
   b. **Recurring Recordings:** Set daily, weekly, or custom pattern recordings
   c. **Program Guide Integration:** Schedule based on program listings
   d. **Priority Settings:** Manage resource allocation for concurrent scheduled recordings
   e. **Notification System:** Alerts for scheduled recording conflicts or failures

5. **Comprehensive Search Functionality**

   Powerful tools to locate specific content:

   a. **Full-Text Search:** Find clips containing specific words in metadata
   b. **Date/Time Search:** Find content from specific broadcast periods
   c. **Channel-Based Filtering:** Focus search on specific channels
   d. **Combined Search Criteria:** Use multiple filters for precision results
   e. **Saved Searches:** Store common search parameters for quick access

6. **Display on Video Wall**

   Video Streaming of 32x IP-TV channels on video wall

7. **Analytics Dashboard**

   Insights into system usage and content patterns:

   a. **Clip Creation Metrics:** Statistics on most frequently clipped content
   b. **Storage Utilization:** Monitoring of system capacity and usage trends
   c. **User Activity Logs:** Tracking of system access and usage patterns
   d. **Content Analysis:** Identification of trending topics or content types
   e. **System Performance Metrics:** Monitoring of recording and processing load

8. **Content Export & Distribution**

   Tools for sharing and utilizing captured content:

   a. **Multi-Format Export:** Convert clips to various video formats
   b. **Batch Export:** Process multiple clips simultaneously
   c. **Scheduled Distribution:** Automate content delivery to designated endpoints
   d. **Access Control:** Manage who can export or distribute specific content

9. **Compliance & Audit Features**

   Tools to support regulatory and internal requirements:

   - **Tamper-Evident Storage:** Ensure recorded content integrity
   - **Access Logs:** Detailed records of who accessed what content and when
   - **Export Tracking:** Complete history of content distribution
   - **Retention Policy Management:** Configurable rules for content preservation
   - **Compliance Reporting:** Generate reports for regulatory requirements

Optional Features

BesTel can offer advanced AI-driven content detection and auto-clipping features as an optional addition to the base system. This would require additional hardware (specifically a GPU) to process the video streams efficiently.

10. **Content Detection & Auto-Clipping (Optional)**

    Advanced AI-driven features for automated content identification:

    a. **Scene Detection:** Automatically identify and clip specific visual content
    b. **Logo/Watermark Recognition:** Detect channel identifiers or program branding
    c. **Text Recognition:** Identify and clip segments containing specific on-screen text
    d. **Audio Cue Detection:** Recognize specific sounds, music, or audio patterns
    e. **Face Recognition:** Identify specific individuals
    f. **Action Detection:** Identify high-motion sequences like sports highlights

g. **Custom Triggers:** User-definable conditions for automated clip creation

Note: Adding this feature would require a GPU (NVIDIA A4000 or similar) to be included in the server configuration. Its price is not included in the current proposal.

Technical Architecture

1. System Deployment

BesTel can offer both on-premises deployment as well as a cloud-based solution. Both these options have their own advantages and disadvantages. However, we were told that the client would prefer an onsite deployment and hence our solution is based on this architecture. Hence, our offer is based on on-premises deployment. of the system This configuration involves installing the server at your location without cloud connectivity. The system operates within your internal network.



Key Aspects

a. Server is installed on Client premises

b. Complete setup for capturing digital streams

c. The services are accessible only within the local network

d. No cloud connectivity required

e. Local management and administration

f. Highspeed and redundant broadband connectivity at the site

g. BesTel shall source the digital stream from a dependable source of its choice

2. Development Stack

Our solution will be built using modern, scalable technologies:

a. **Frontend:** React.js for a responsive and intuitive user interface

b. **Backend:** Node.js for API services and system management

c. **Processing Engine:** Python for video processing and AI functions

d. **Database:** PostgreSQL for metadata storage and search indexing

e. **Storage:** Custom file system management for optimized video storage

3. **Hardware Requirements - Minimum**

CMC to provide the following hardware for the project:

a. **Server (Qty 1)**

- **Model:** Dell PowerEdge 740
- **CPU:** Intel Xeon Silver 4210R (10 cores, 20 threads)
- **RAM:** 128GB DDR4 ECC Memory
- **System Storage:** 500GB NVMe SSD in RAID 1 for OS and applications
- **Recording Storage (See Annex 1)**
  - Primary: 6TB x 14 (84TB) Dell Enterprise Plus 3.5 HDD SAS 12 Gbps, 7.2K RPM for 84TB Raw Storage + 6TB x 2 Spare HDDs on standby
  - Configuration: RAID 6 with hot spares
  - Usable: 72TB after RAID 6
- **Network (See Annex 2)**
  - 1 x Quad Port 10G SFP+ Network Card with SFP Modules
  - 1 x Dual Port 1GB Base-T RJ-45 PCI-e
- **Power:** Redundant 1200W Power Supplies
- **Form Factor:** 4U Rack-mountable chassis

b. **Network Switch (Qty 1)**

- 8-Port Gigabit Smart Switch with 2 SFP Slots

c. **Desktop Computer (Qty 1)**

- Dell OptiPlex 980 - Intel Core i5, 16GB, 512GB SSD, 2xHDMI ports

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Project Deliverables

- **Core Recording Engine** - Base recording functionality
- **Storage Management System** - Archive and retrieval system
- **Clip Management Features** - Clip creation and organization tools
- **AI Detection System** - Content recognition and auto-clipping
- **User Interface Development** - Complete frontend and control panel
- **Testing & Optimization** - QA, performance tuning, stress testing
- **Deployment & Installation** - Hardware setup and software deployment
- **Training & Handover** - Staff training and documentation

Project Duration

- **8 weeks** from the signing of the agreement.

Terms & Conditions

a. **Scope**: This quotation is valid only for digital stream input. Any additional requirements for analog signal processing or conversion will incur additional costs.

b. **Validity**: This proposal is valid for 30 days from the date of issue.

c. **Confidentiality**: Both parties agree to maintain the confidentiality of all information shared during the project.

d. **Project Changes**: Any changes to the scope of work after project initiation may result in additional costs and timeline adjustments.

BesTel looks forward to partnering with CMC to deliver this advanced IPTV Recording & Management System that will transform Client's content management capabilities.



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Annex-1

IPTV Storage & Network Requirements

Storage Requirements Summary

For 32 IPTV channels at 1080p resolution and 25 fps recorded 24/7 for 30 days:

Encoding	Bitrate	Storage Required
H.264/AVC	4096 Kbps (4 Mbps)	38.7 TB
H.264/AVC	5 Mbps	48.3 TB
H.265/HEVC	2.5 Mbps	24.1 TB
MPEG-2	8 Mbps	77.3 TB
MPEG-4	6 Mbps	58.0 TB

Calculation Method

Storage (TB) = Bitrate (Mbps) × Hours × Channels × 0.000419

Example: 4 Mbps × (30 × 24 hours) × 32 channels × 0.000419 = 38.7 TB

Common IPTV Encoding Formats

Encoding Format	Typical Bitrate (1080p/25fps)	Advantages	Disadvantages
H.264/AVC	4-5 Mbps	Industry standard, widely supported	Higher storage needs than newer codecs
H.265/HEVC	2-3 Mbps	40-50% better compression than H.264	Higher processing requirements
MPEG-2	6-8 Mbps	Legacy compatibility	Inefficient compression
MPEG-4 Part 2	5-6 Mbps	Older equipment compatibility	Less efficient than H.264
AV1	1.5-2.5 Mbps	Most efficient compression	Highest processing requirements

Recommendation

A total usable storage capacity of 72TB is recommended for Analysys software, video clips, images and database. We would implement RAID 6.

Annex 2

Network Bandwidth Requirement

Bandwidth Calculation

32 channels at 4096 Kbps (4 Mbps) per channel requires:

32 × 4 Mbps = 128 Mbps (baseline bandwidth)

For 5 Mbps streams:

32 x 5 Mbps = 160 Mbps (baseline bandwidth)

1G Network Card

The baseline calculation shows 128-160 Mbps theoretical bandwidth requirement, hence we recommend a 1G network card.

A 1G network interface will provide the necessary headroom to ensure a reliable, high-quality recording of all 32 channels simultaneously, even during peak bandwidth events.

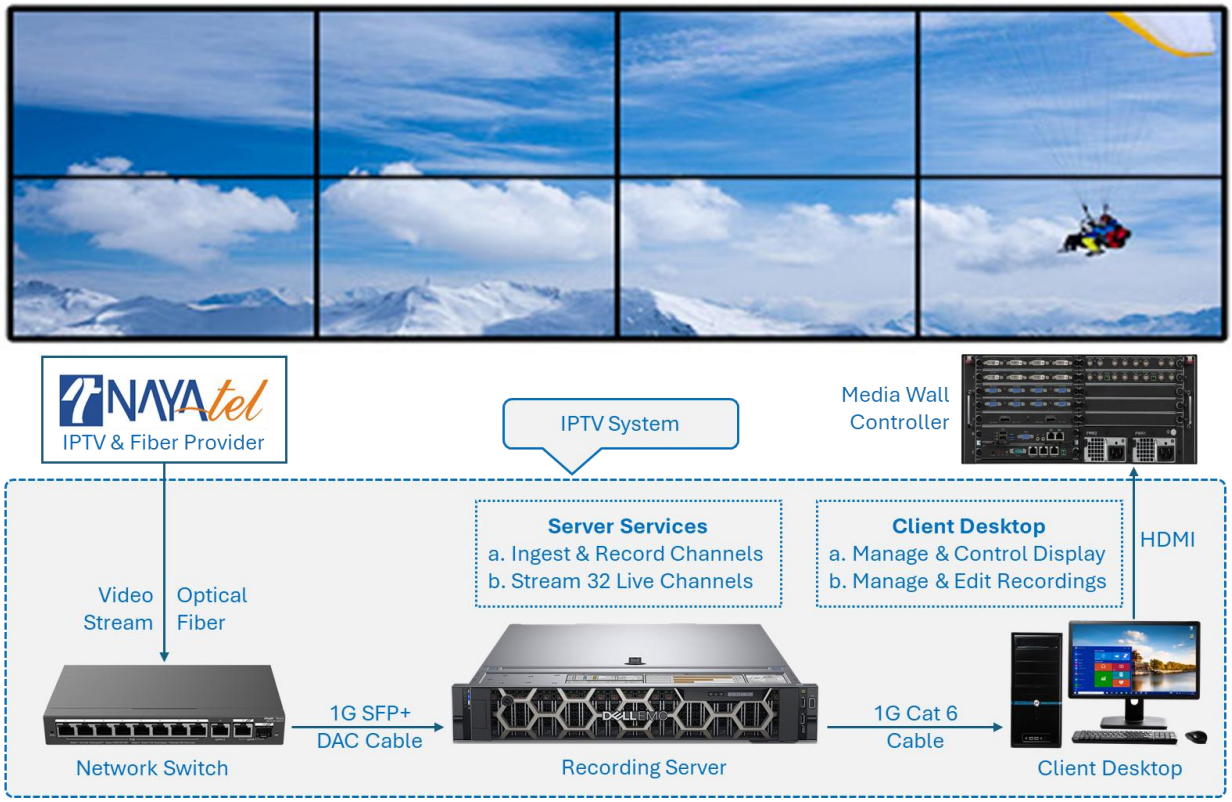
Recommendation

- **Network capacity**: 1G network interface
- **Encoding recommendation**: H.264/AVC at 4Mbps/ 5Mbps (industry standard)

Annex 3

Network High Level Diagram

High-Level Design (HLD) for Solution Integration



The solution will integrate seamlessly with your existing network infrastructure and the video wall.

a. Media Stream

CMC will arrange streams of the desired 32 IPTV channels from Nayatel over an optical fiber link to the Recording Server. This channel / video stream can be either on HLS or UDP formats.

b. BesTel System

Network Switch will be used for the termination of channel streams from Nayatel. We might be able to terminate the stream directly on the server, eliminating a potential point of failure.

Media Server will run BesTel's custom-built media management software, Analysys, and stream recordings and live streams to the Client Desktop. The Media Server will connect to the network switch via the high-performance 1G SFP+ DAC Cable.

Analysys is BesTel's media management software that will ingest, record and manage recordings and the live channels.

Client Desktop will display live streams and recordings onto the Video Wall. The Client Desktop will have multiple channel projection templates to display the channels in the desired format.

c. Video Wall

Controller will be connected to the Client Desktop via an HDMI interface.

Annex-4

Bill of Quantities (BoQ)

BoQ for BesTel Recording Solution

The proposed IPTV Management and Monitoring Solution will comprise of the following equipment, devices, software and services under this proposal:

No.	Company	Items	Specifications	Qty
1.	BesTel	Media Management Software	Analysys Media Management Software - Digital TV channel ingestion, transcoding, recording, clipping, repurposing, storing, indexing, retrieving, displaying.	1
2.	CMC	Switch	8-Port Gigabit Smart Network Switch with 2 SFP Slots and matching SFPs	1
3.	CMC	Media and Recording Server	Dell PowerEdge 740 Intel Xeon Silver 4210R (10 cores, 20 threads), 128GB DDR4 ECC Memory, 500GB NVMe SSD, 6TB x 14 (84TB) Dell Enterprise Plus 3.5 HDD SAS 12 Gbps, 7.2K RPM for 84TB Raw Storage, Quad Port 1G SFP+ Network Card, 2 Ports 1GB Base-T RJ-45 PCI-e, Redundant Power Supplies, 4U Rack-mountable chassis, Monitor, Keyboard, Mouse	1
4.	CMC	Client Desktop	Dell OptiPlex 980 - Intel Core i5, 16GB, 512GB SSD, 2 x HDMI, 21” Monitor, Keyboard, Mouse	1
5.	CMC	Cables and Accessories	Patch cords, HDMI cables, cable ties, connectors, etc.	Set
6.	BesTel	Services	Installation, Configuration, Testing, Commissioning	Job

a. Notes

- Analysys license is perpetual.
- Any additional feature required by the client in Analysys would be quoted separately.
- Specifications of network switch, server, and the desktop are provided in the above.

b. Software Support and Maintenance Services

Our proposal includes One (1) year of Software Support options. The software support and maintenance services would comprise of remote support and troubleshooting with onsite assistance from CMC technical staff.

Annex 5

Server Specifications Assessment

CPU Sufficiency Assessment

The Intel Xeon Silver 4210R (10 cores, 20 threads) CPU is sufficient for 32-channel recording with the following considerations:

- For basic recording functionality, the CPU provides adequate processing power
- The CPU utilization will be approximately 60-70% during normal operation
- The system will function well for recording and basic playback functions
- For optimal performance during resource-intensive operations (multi-channel simultaneous viewing, advanced search, or export), a GPU would improve performance but is not mandatory

```
CPU: 10 cores, 20 threads (Intel Xeon Silver 4210R)
Estimated CPU utilization: 240.00% (7.5% per stream per core and we have 10 cores)
Sufficiency rating: Good

RAM: 64GB available
Estimated RAM needed: 9.38GB (300MB per stream)
Sufficiency rating: Excellent

Network: 4×10G SFP+ ports (5000MB/s)
Estimated network throughput: 16.00MB/s (128Mbps)
Sufficiency rating: Excellent

Storage: 192TB usable
Estimated storage for 30 DAYS: 39.55TB to 48.5TB
Maximum retention period possible: 146 days
Sufficiency rating: Very good
```

GPU Requirements Assessment

The solution **can function without a GPU** to meet your specifications and 30-day retention requirement. However, there are important considerations:

- Without GPU acceleration, all video processing relies on CPU resources
- Basic recording functions will operate properly without GPU
- Simultaneous viewing of multiple channels may experience reduced performance
- If future expansion or advanced analytics features are planned, adding a GPU would be recommended

Annex 6

Video Format & Data Sizing

Video Format Specifications

The recording solution will:

- Record in industry-standard H.264/H.265 formats
- No additional encoding/decoding is needed if the IPTV streams already use these formats
- If the IPTV streams use different formats, the system will transcode them appropriately

Exportable Formats

These would include:

- MP4 (most compatible format)
- AVI (Windows-compatible)
- MKV (high quality container)
- TS (Transport Stream - original broadcast format)

Export and Editing Compatibility

After exporting, recordings can be edited using our own software. There is no need to use any other software or tool.

Data Size Estimation

Based on detailed calculations for 1080p H.264 streams at standard quality (4 Mbps):

- **Per channel:** 42.19 GB per day
- **All 32 channels:** 1.32 TB per day
- **30-day retention total:** 39.55 TB ~ 48.5TB

But we need high flexibility in the storage

Annex 7

BesTel Responsibilities

BesTel Responsibility for Recording

BesTel will take full responsibility for ensuring all 32 channels record continuously for the required 30-day period. This includes:

- Configuration of all recording channels
- Monitoring tools to verify recording status in real-time
- Automated notification system for recording failures
- Regular verification of recording integrity
- Technical support for any recording issues

Analysys is designed with redundancy features to ensure no data loss, and our system will actively monitor its performance.

Software Support and Hardware Warranty

Our proposal is inclusive of one year software support which would cover:

- Remote support for Analysys functionality and performance
- Remote Technical support and troubleshooting of Analysys
- System updates and security patches

In case of server/desktop crash we shall:

- Provide data recovery services where possible
- Extend remote and on-site technical support, as needed

Installation and Setup

Our quotation / charges are inclusive of:

- Complete Analysys software deployment
- Configuration of all 32 channels for recording
- Integration with the existing network infrastructure
- System documentation and user manuals
- Testing and validation of all system functions
- Commissioning of the system and go-live

Annex 9

Proposed HDD Make and Model

The **Dell 6TB SAS 7.2k 3.5" 12G 4Kn Enterprise Class** Hard Drive (R69WP) is designed for demanding IT infrastructures, offering high capacity and performance. Key features include:

- **Capacity:** 6TB
- **Interface:** SAS 12Gb/s
- **Spindle Speed:** 7200 RPM
- **Form Factor:** 3.5"
- **Data Transfer Rate:** 1200 Mbps (external)
- **Sector Size:** 4Kn
- **Features:** Enterprise Class, 4Kn



This drive is suitable for businesses needing high-capacity storage and robust data management, and it's designed to integrate seamlessly with Dell PowerEdge servers and storage systems.